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Bereken in $\mathbb{C}$ de vierkantswortel van het volgende complexe getal: $-1 - 2\sqrt{2}i$.

$$-1 - 2\sqrt{2}i = (x + yi)^2 = x^2 - y^2 + 2xyi$$

$$\Leftrightarrow \begin{cases} x^2 - y^2 = -1 \\ 2xy = -2\sqrt{2} \end{cases}$$

$$\Leftrightarrow \begin{cases} x^2 - y^2 = -1 \\ xy = -\sqrt{2} \end{cases}$$

$$\Rightarrow \begin{cases} x^2 - y^2 = -1 \\ x^2 y^2 = 2 \end{cases}$$

$$\Leftrightarrow \begin{cases} x^2 + 1 = y^2 \\ x^2(x^2 + 1) = 2 \end{cases}$$

$$\Leftrightarrow \begin{cases} x^2 + 1 = y^2 \\ x^4 + x^2 - 2 = 0 \end{cases}$$

Laat $x^2 = \lambda$

$$\lambda^2 + \lambda - 2 = 0$$

$$(\lambda + 2)(\lambda - 1) = 0$$

$$\lambda_1 = -2 \text{ en } \lambda_2 = 1$$

$$x = 1 \text{ of } x = -1 \text{ en } y = \sqrt{2} \text{ of } y = -\sqrt{2}$$

Aangezien $xy = -\sqrt{2}$, dan hebben x en y tegengesteld teken

Oplossing: $1 - \sqrt{2}i$ of $-1 + \sqrt{2}i$

References